GB_FinalProject

GB

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1 Loading all required libraries and set chunk options

2 Loading the dataset

```r
mortality <- read.csv("Mortality.csv")
head(mortality)
```

```
##   Treatment Replication initialJ2 finalJ2 Mortality
## 1         1           A        55      40        27
## 2         1           B        42      32        24
## 3         1           C        57      45        21
## 4         1           D        53      38        28
## 5         2           A        58      57         2
## 6         2           B        62      60         3
```

3 Calculate percent mortality for each observation

$$((InitialJ2 - FinalJ2)/InitialJ2) * 100$$

```
Mortality <- mortality %>%
   mutate(Mortality_percent = ((initialJ2 - finalJ2) / initialJ2) * 100)
head(Mortality)
```

```
##   Treatment Replication initialJ2 finalJ2 Mortality Mortality_percent
## 1         1           A        55      40        27         27.272727
## 2         1           B        42      32        24         23.809524
## 3         1           C        57      45        21         21.052632
## 4         1           D        53      38        28         28.301887
## 5         2           A        58      57         2          1.724138
## 6         2           B        62      60         3          3.225806
```

4 Summary Statistics by Treatment

```
summary_stats <- Mortality %>%
  group_by(Treatment) %>% # Group by treatment and calculate mean, SD, SE
  summarise(n = n(),
    mean_mortality = mean(Mortality_percent),
            sd_mortality = sd(Mortality_percent),
            se = sd_mortality/n )

# # Display as table

knitr::kable(summary_stats)
```

Treatment	n	mean_mortality	sd_mortality	se
1	4	25.1091924	3.3175797	0.8293949
10	4	2.5816993	3.0383371	0.7595843
100	4	4.9921558	0.4028120	0.1007030
101	4	3.3828268	0.8782267	0.2195567
103	4	26.3644145	7.1159648	1.7789912
104	4	1.4361300	1.7918775	0.4479694
105	4	8.9801076	3.1710824	0.7927706
106	4	1.4474773	1.6930926	0.4232731
107	4	7.9320898	3.9724387	0.9931097
108	4	4.9603175	1.9841270	0.4960317
11	4	8.8983382	3.0459402	0.7614850
110	4	3.8629997	1.9428747	0.4857187
111	4	9.0992130	3.2346848	0.8086712
112	4	2.6685192	1.2705643	0.3176411
113	4	2.5902778	0.9946811	0.2486703
114	4	0.5813953	1.1627907	0.2906977
115	4	1.9399460	2.4610386	0.6152597
117	4	3.5086529	1.2011010	0.3002752
118	4	5.9278455	1.7362320	0.4340580

Treatment	n	mean_mortality	sd_mortality	se
119	4	1.1858798	1.3957423	0.3489356
12	4	1.7725468	2.2371460	0.5592865
121	4	9.6026184	3.0888215	0.7722054
122	4	14.2872702	6.6201412	1.6550353
123	4	9.6585529	1.8568843	0.4642211
124	4	9.5090228	3.9682654	0.9920664
125	4	11.4074476	3.5934665	0.8983666
126	4	9.4024032	1.3729289	0.3432322
128	4	1.7917471	1.2209809	0.3052452
129	4	9.5699878	2.4776617	0.6194154
13	4	2.1083376	1.5298068	0.3824517
130	4	11.1590822	2.5347321	0.6336830
131	4	1.2989204	1.5001170	0.3750293
133	4	4.5961667	1.5863485	0.3965871
134	4	10.0186921	1.2127741	0.3031935
135	4	0.5000000	1.0000000	0.2500000
136	4	9.2776825	2.7377640	0.6844410
137	4	13.3167167	4.5793263	1.1448316
138	4	8.4853178	1.4522188	0.3630547
139	4	9.2536073	4.0818808	1.0204702
14	4	18.2964641	8.9161808	2.2290452
141	4	3.8561637	1.9645714	0.4911429
143	4	1.8292683	3.6585366	0.9146341
146	4	5.6344413	0.7862776	0.1965694
147	4	1.2740184	1.6671848	0.4167962
148	4	2.5838745	2.2334854	0.5583714
149	4	7.6541738	2.1386141	0.5346535
15	4	3.0808559	1.4207591	0.3551898
150	4	2.5248334	1.4995098	0.3748774
151	4	5.8294101	1.3689943	0.3422486
152	4	12.2532006	3.5165918	0.8791479
153	4	4.4846813	3.5949425	0.8987356
154	4	4.5794953	2.5494969	0.6373742
155	4	4.8174307	3.0359940	0.7589985
156	4	2.6688230	0.6745875	0.1686469
157	4	12.2945605	5.3504082	1.3376020
158	4	1.0110294	1.1685077	0.2921269
159	4	2.1515560	0.7015917	0.1753979
16	4	4.7667254	1.4975575	0.3743894
160	4	3.3410161	3.3070073	0.8267518
162	4	1.2956811	1.5117800	0.3779450
163	4	3.2278139	3.1319079	0.7829770
164	4	1.0621646	1.2374408	0.3093602
165	4	1.6160714	1.0897193	0.2724298
166	4	0.9433962	1.8867925	0.4716981
167	4	6.1681984	1.4790103	0.3697526
168	4	4.1005291	4.8130702	1.2032676
169	4	3.7211740	5.2379445	1.3094861
17	4	5.3516814	0.2126431	0.0531608
170	4	4.7035588	1.7472976	0.4368244
171	4	2.0068685	1.6715622	0.4178906
172	4	3.2866379	3.8042507	0.9510627

Treatment	n	mean_mortality	sd_mortality	se
174	4	2.5273460	2.0329365	0.5082341
175	4	1.3782051	1.8232644	0.4558161
176	4	7.6656315	4.6877039	1.1719260
177	4	2.0528455	2.4528740	0.6132185
178	4	2.3065476	3.3704618	0.8426155
18	4	5.5511693	1.0208844	0.2552211
180	4	5.2421846	2.7487254	0.6871814
181	4	1.9879924	1.5621293	0.3905323
182	4	1.9261174	1.8015298	0.4503824
183	4	2.1906146	3.2999275	0.8249819
184	4	3.3383608	1.1713865	0.2928466
185	4	1.7602041	2.3644349	0.5911087
187	4	1.5776503	1.1033511	0.2758378
188	4	2.6346522	1.9333692	0.4833423
189	4	8.0117719	2.0890987	0.5222747
19	4	8.5175887	1.0557815	0.2639454
190	4	5.0353938	2.5728626	0.6432156
191	4	0.5813953	1.1627907	0.2906977
192	4	1.8951879	2.3617094	0.5904273
193	4	1.7873652	2.5720359	0.6430090
194	4	8.4643858	1.4694663	0.3673666
195	4	3.9613238	3.9980893	0.9995223
196	4	1.5531523	1.8743581	0.4685895
197	4	0.0000000	0.0000000	0.0000000
198	4	2.4142415	1.8561115	0.4640279
199	4	1.1022287	1.2765806	0.3191452
2	4	4.4324013	2.3634407	0.5908602
20	4	6.3726393	3.4480979	0.8620245
200	4	2.0623249	1.7021417	0.4255354
201	4	1.8911800	0.9532463	0.2383116
203	4	7.5813216	1.7388469	0.4347117
204	4	32.3644332	10.6012108	2.6503027
206	4	27.1478155	19.4864196	4.8716049
207	4	4.4091494	0.9272724	0.2318181
208	4	7.4308883	0.8962024	0.2240506
209	4	3.3178343	1.6501973	0.4125493
21	4	3.2043309	2.9043038	0.7260760
210	4	5.6931573	1.5348211	0.3837053
211	4	15.2422438	2.5460160	0.6365040
212	4	5.1896552	2.0069376	0.5017344
213	4	4.1773241	1.0309346	0.2577337
214	4	2.7701201	1.1921280	0.2980320
215	4	2.8198208	0.6170789	0.1542697
216	4	3.0659524	2.2661711	0.5665428
217	4	2.0235115	0.5746991	0.1436748
218	4	3.1356275	0.6919919	0.1729980
219	4	2.6339628	1.0993738	0.2748435
22	4	8.5074557	2.9433789	0.7358447
220	4	2.6316102	0.8577608	0.2144402
221	4	2.7242195	1.1081330	0.2770333
222	4	1.4109236	0.4861677	0.1215419
223	8	8.2411925	2.0017132	0.2502141

Treatment	n	mean_mortality	sd_mortality	se
224	4	2.2665889	0.8921439	0.2230360
225	4	2.5470326	1.1243052	0.2810763
226	4	1.9440395	0.5691732	0.1422933
227	4	2.3172859	0.7623444	0.1905861
228	4	2.4082987	0.4062139	0.1015535
229	4	2.7093099	0.8358153	0.2089538
23	4	4.7120693	2.7365880	0.6841470
230	4	3.1670140	0.7810223	0.1952556
231	4	9.4987871	3.1499282	0.7874820
234	4	7.5768699	1.1104142	0.2776036
235	4	21.0207937	6.7576148	1.6894037
238	4	6.5595441	1.6693623	0.4173406
239	4	20.2462070	5.7303655	1.4325914
24	4	3.7244076	1.5116585	0.3779146
240	4	13.9150514	9.8780088	2.4695022
243	4	6.4390140	1.8294692	0.4573673
245	4	14.5434441	3.7570036	0.9392509
246	4	11.2559181	5.8108064	1.4527016
247	4	15.5749543	8.8903490	2.2225872
249	4	13.9280495	4.6095135	1.1523784
25	4	3.6058718	1.6167512	0.4041878
26	4	2.8368193	1.2675684	0.3168921
27	4	8.9219530	5.1776064	1.2944016
28	4	10.5672515	2.1540481	0.5385120
29	4	16.3073461	9.2418717	2.3104679
3	4	36.0490271	12.7022994	3.1755748
30	4	12.5954803	3.9666382	0.9916596
32	4	11.4694354	3.4286442	0.8571611
34	4	7.9634610	1.8724632	0.4681158
36	4	29.2489496	14.7087870	3.6771967
38	4	10.5942552	3.5770161	0.8942540
4	4	4.0087866	2.1483203	0.5370801
41	4	9.5738326	2.1801033	0.5450258
42	4	15.4191033	5.7074161	1.4268540
43	4	20.9392084	4.1116350	1.0279087
45	4	8.5991603	2.4830832	0.6207708
46	4	3.6067010	1.9081120	0.4770280
47	4	2.6495123	1.6552084	0.4138021
48	4	9.5279974	1.5778912	0.3944728
49	4	2.5189988	2.0424901	0.5106225
5	4	5.6692799	2.6157530	0.6539382
50	4	27.2321429	13.6288728	3.4072182
52	4	10.1554327	3.4320788	0.8580197
53	4	22.2836865	7.9124786	1.9781196
54	4	26.1465473	4.8986381	1.2246595
55	4	16.7748285	6.2621909	1.5655477
56	4	18.2668105	3.3553544	0.8388386
57	4	19.2645719	7.8874578	1.9718645
58	4	4.2249134	0.8074494	0.2018623
59	8	25.6644889	10.9460389	1.3682549
6	4	6.0176651	1.8189798	0.4547449
61	4	14.2201599	6.0950177	1.5237544

Treatment	n	mean_mortality	sd_mortality	se
62	4	14.6098369	6.4795508	1.6198877
63	4	17.2959862	1.8463834	0.4615958
64	4	30.8831522	4.6321906	1.1580477
65	4	23.7000440	10.4549022	2.6137255
66	4	19.1174948	8.2996842	2.0749210
67	4	9.8789620	4.3837011	1.0959253
68	4	19.5015534	15.2697323	3.8174331
69	4	23.2183212	7.0716839	1.7679210
7	4	6.7112135	3.6797055	0.9199264
70	4	12.3920524	9.9356933	2.4839233
71	4	14.1596639	2.9401981	0.7350495
72	4	11.4028857	4.1760262	1.0440065
73	4	8.4585295	1.4789508	0.3697377
74	4	6.8038926	0.7646907	0.1911727
75	4	5.3902625	2.5710777	0.6427694
76	4	6.3902316	2.5148728	0.6287182
77	4	8.2542870	2.9074548	0.7268637
78	4	3.6092422	2.2271856	0.5567964
79	4	1.3059163	1.7244320	0.4311080
8	4	5.3350324	2.4378031	0.6094508
80	4	7.7498283	1.9765973	0.4941493
81	4	11.9318182	1.9483225	0.4870806
82	4	9.0699118	3.4232567	0.8558142
83	4	8.7801867	1.9028644	0.4757161
84	4	5.3656951	2.7220482	0.6805120
85	4	4.5746214	2.0968625	0.5242156
86	4	12.5177900	4.6386538	1.1596635
87	4	10.3797992	5.4963435	1.3740859
88	4	9.1331881	3.0989268	0.7747317
89	4	9.9914185	3.8391062	0.9597766
9	4	11.4737365	4.6911821	1.1727955
90	4	5.5429515	0.8792379	0.2198095
91	4	5.8369643	2.5305802	0.6326451
92	4	3.7189377	2.8234380	0.7058595
93	4	8.6864913	2.8717560	0.7179390
94	4	5.1540671	1.6207336	0.4051834
95	4	3.7243394	1.5644764	0.3911191
97	4	4.5333588	1.6735223	0.4183806
98	4	2.8328032	2.1509457	0.5377364
99	4	5.2083333	2.6672513	0.6668128
Fluopyram	4	98.5694444	1.1550346	0.2887587
UntreatedControl	4	1.4391447	1.6739457	0.4184864

5 selecting Top Treatments with Mean mortality higher than 50% and untreated control

```
# Filter strains with >50% mortality or named 'UntreatedControl'
selected_strains <- summary_stats %>%
```

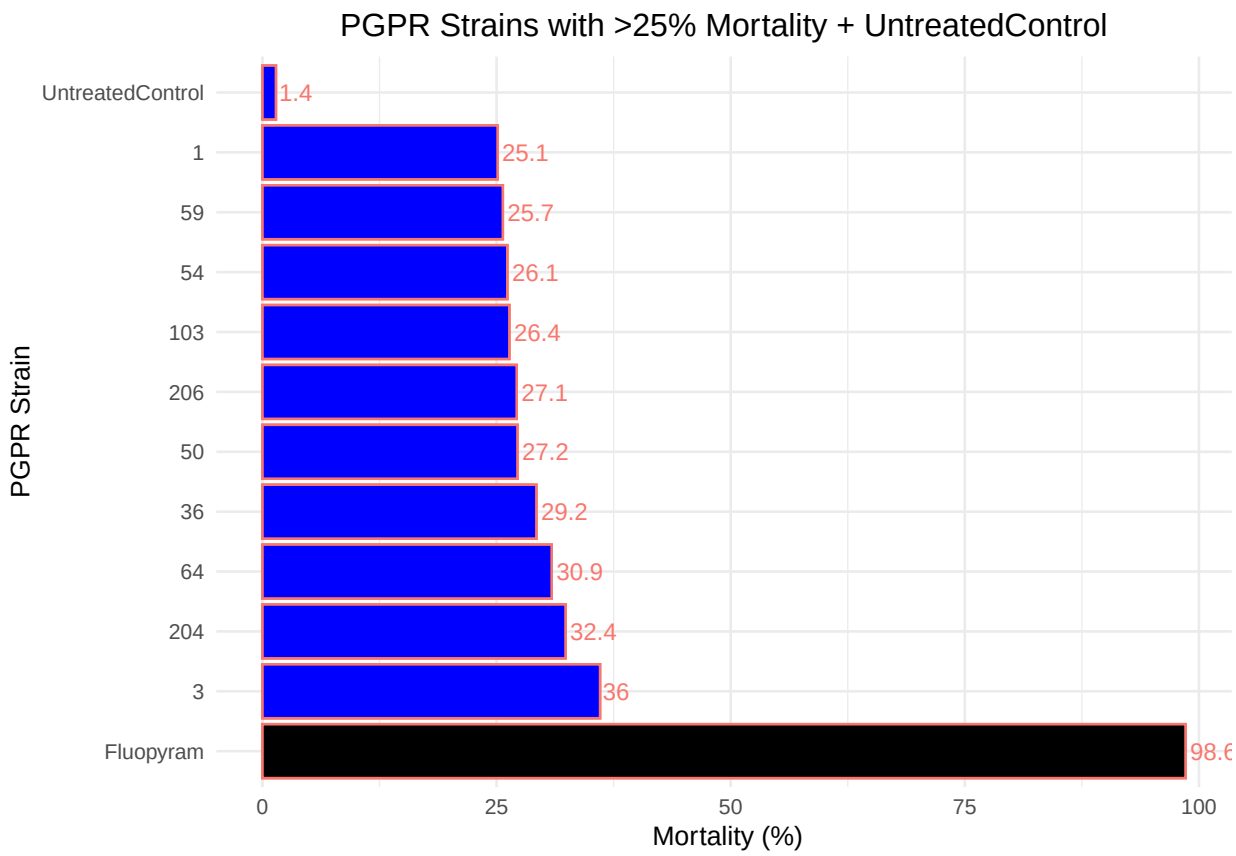
```r
  filter(mean_mortality > 25 | Treatment == "UntreatedControl")
selected_strains
```

```
## # A tibble: 12 x 5
##    Treatment            n mean_mortality sd_mortality    se
##    <chr>            <int>          <dbl>        <dbl> <dbl>
##  1 1                    4          25.1          3.32 0.829
##  2 103                  4          26.4          7.12 1.78
##  3 204                  4          32.4         10.6  2.65
##  4 206                  4          27.1         19.5  4.87
##  5 3                    4          36.0         12.7  3.18
##  6 36                   4          29.2         14.7  3.68
##  7 50                   4          27.2         13.6  3.41
##  8 54                   4          26.1          4.90 1.22
##  9 59                   8          25.7         10.9  1.37
## 10 64                   4          30.9          4.63 1.16
## 11 Fluopyram            4          98.6          1.16 0.289
## 12 UntreatedControl     4           1.44         1.67 0.418
```

6 Plot Manuscript-Ready Figure

```r
# Barplot of selected treatments, color-coded with red for Untreatedcontrol and green for Fluopyram
ggplot(selected_strains, aes(x = reorder(Treatment, -mean_mortality), y = mean_mortality, color="black")
  geom_col(aes(fill = Treatment == "UntreatedControl")) +
   scale_fill_manual(values = c("TRUE" = "red", "FALSE" = "blue")) +
  geom_col(aes(fill = Treatment == "Fluopyram")) +
  scale_fill_manual(values = c("TRUE" = "Black", "FALSE" = "blue")) +
  geom_text(aes(label = round(mean_mortality, 1)), hjust = -0.1, size = 3) +
  coord_flip() +
  ylab(" Mortality (%)") +
  xlab("PGPR Strain") +
  ggtitle("PGPR Strains with >25% Mortality + UntreatedControl") +
   theme_minimal(base_size = 10) +
  theme(plot.title = element_text(hjust = 0.5),
        legend.position = "none")
```

```
## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.
```



7 Mixed Effects Model (GLMM)

```
# Convert grouping variables to factors
Mortality$Treatment <- as.factor(Mortality$Treatment)
Mortality$Replication <- as.factor(Mortality$Replication)
```

```
# Linear mixed model with rep as random effect
mod <- lmer(Mortality_percent  ~ Treatment + (1|Replication),
            data = Mortality)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Summary of the model, includes fixed effects and random structure
summary(mod)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Mortality_percent ~ Treatment + (1 | Replication)
##    Data: Mortality
##
## REML criterion at convergence: 4070.9
##
```

```
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9794 -0.3610 -0.0422  0.3404  6.1679
##
## Random effects:
##  Groups      Name        Variance  Std.Dev.
##  Replication (Intercept) 7.482e-13 8.650e-07
##  Residual                1.783e+01 4.222e+00
## Number of obs: 876, groups:  Replication, 4
##
## Fixed effects:
##                          Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)               25.1092     2.1112 659.0000  11.893  < 2e-16 ***
## Treatment10              -22.5275     2.9857 659.0000  -7.545 1.51e-13 ***
## Treatment100             -20.1170     2.9857 659.0000  -6.738 3.51e-11 ***
## Treatment101             -21.7264     2.9857 659.0000  -7.277 9.76e-13 ***
## Treatment103               1.2552     2.9857 659.0000   0.420 0.674327
## Treatment104             -23.6731     2.9857 659.0000  -7.929 9.49e-15 ***
## Treatment105             -16.1291     2.9857 659.0000  -5.402 9.21e-08 ***
## Treatment106             -23.6617     2.9857 659.0000  -7.925 9.76e-15 ***
## Treatment107             -17.1771     2.9857 659.0000  -5.753 1.34e-08 ***
## Treatment108             -20.1489     2.9857 659.0000  -6.748 3.28e-11 ***
## Treatment11              -16.2109     2.9857 659.0000  -5.429 7.96e-08 ***
## Treatment110             -21.2462     2.9857 659.0000  -7.116 2.91e-12 ***
## Treatment111             -16.0100     2.9857 659.0000  -5.362 1.14e-07 ***
## Treatment112             -22.4407     2.9857 659.0000  -7.516 1.85e-13 ***
## Treatment113             -22.5189     2.9857 659.0000  -7.542 1.54e-13 ***
## Treatment114             -24.5278     2.9857 659.0000  -8.215 1.12e-15 ***
## Treatment115             -23.1692     2.9857 659.0000  -7.760 3.25e-14 ***
## Treatment117             -21.6005     2.9857 659.0000  -7.235 1.30e-12 ***
## Treatment118             -19.1813     2.9857 659.0000  -6.424 2.54e-10 ***
## Treatment119             -23.9233     2.9857 659.0000  -8.013 5.11e-15 ***
## Treatment12              -23.3366     2.9857 659.0000  -7.816 2.16e-14 ***
## Treatment121             -15.5066     2.9857 659.0000  -5.194 2.75e-07 ***
## Treatment122             -10.8219     2.9857 659.0000  -3.625 0.000312 ***
## Treatment123             -15.4506     2.9857 659.0000  -5.175 3.03e-07 ***
## Treatment124             -15.6002     2.9857 659.0000  -5.225 2.34e-07 ***
## Treatment125             -13.7017     2.9857 659.0000  -4.589 5.33e-06 ***
## Treatment126             -15.7068     2.9857 659.0000  -5.261 1.94e-07 ***
## Treatment128             -23.3174     2.9857 659.0000  -7.810 2.27e-14 ***
## Treatment129             -15.5392     2.9857 659.0000  -5.204 2.60e-07 ***
## Treatment13              -23.0009     2.9857 659.0000  -7.704 4.87e-14 ***
## Treatment130             -13.9501     2.9857 659.0000  -4.672 3.61e-06 ***
## Treatment131             -23.8103     2.9857 659.0000  -7.975 6.77e-15 ***
## Treatment133             -20.5130     2.9857 659.0000  -6.870 1.49e-11 ***
## Treatment134             -15.0905     2.9857 659.0000  -5.054 5.61e-07 ***
## Treatment135             -24.6092     2.9857 659.0000  -8.242 9.14e-16 ***
## Treatment136             -15.8315     2.9857 659.0000  -5.302 1.56e-07 ***
## Treatment137             -11.7925     2.9857 659.0000  -3.950 8.67e-05 ***
## Treatment138             -16.6239     2.9857 659.0000  -5.568 3.76e-08 ***
## Treatment139             -15.8556     2.9857 659.0000  -5.310 1.50e-07 ***
## Treatment14               -6.8127     2.9857 659.0000  -2.282 0.022822 *
## Treatment141             -21.2530     2.9857 659.0000  -7.118 2.87e-12 ***
## Treatment143             -23.2799     2.9857 659.0000  -7.797 2.48e-14 ***
```

```
## Treatment146          -19.4748     2.9857 659.0000  -6.523 1.38e-10 ***
## Treatment147          -23.8352     2.9857 659.0000  -7.983 6.36e-15 ***
## Treatment148          -22.5253     2.9857 659.0000  -7.544 1.52e-13 ***
## Treatment149          -17.4550     2.9857 659.0000  -5.846 7.92e-09 ***
## Treatment15           -22.0283     2.9857 659.0000  -7.378 4.86e-13 ***
## Treatment150          -22.5844     2.9857 659.0000  -7.564 1.32e-13 ***
## Treatment151          -19.2798     2.9857 659.0000  -6.457 2.07e-10 ***
## Treatment152          -12.8560     2.9857 659.0000  -4.306 1.92e-05 ***
## Treatment153          -20.6245     2.9857 659.0000  -6.908 1.16e-11 ***
## Treatment154          -20.5297     2.9857 659.0000  -6.876 1.43e-11 ***
## Treatment155          -20.2918     2.9857 659.0000  -6.796 2.41e-11 ***
## Treatment156          -22.4404     2.9857 659.0000  -7.516 1.85e-13 ***
## Treatment157          -12.8146     2.9857 659.0000  -4.292 2.04e-05 ***
## Treatment158          -24.0982     2.9857 659.0000  -8.071 3.31e-15 ***
## Treatment159          -22.9576     2.9857 659.0000  -7.689 5.41e-14 ***
## Treatment16           -20.3425     2.9857 659.0000  -6.813 2.16e-11 ***
## Treatment160          -21.7682     2.9857 659.0000  -7.291 8.86e-13 ***
## Treatment162          -23.8135     2.9857 659.0000  -7.976 6.71e-15 ***
## Treatment163          -21.8814     2.9857 659.0000  -7.329 6.83e-13 ***
## Treatment164          -24.0470     2.9857 659.0000  -8.054 3.76e-15 ***
## Treatment165          -23.4931     2.9857 659.0000  -7.868 1.48e-14 ***
## Treatment166          -24.1658     2.9857 659.0000  -8.094 2.79e-15 ***
## Treatment167          -18.9410     2.9857 659.0000  -6.344 4.17e-10 ***
## Treatment168          -21.0087     2.9857 659.0000  -7.036 4.96e-12 ***
## Treatment169          -21.3880     2.9857 659.0000  -7.163 2.11e-12 ***
## Treatment17           -19.7575     2.9857 659.0000  -6.617 7.58e-11 ***
## Treatment170          -20.4056     2.9857 659.0000  -6.834 1.88e-11 ***
## Treatment171          -23.1023     2.9857 659.0000  -7.738 3.82e-14 ***
## Treatment172          -21.8226     2.9857 659.0000  -7.309 7.82e-13 ***
## Treatment174          -22.5818     2.9857 659.0000  -7.563 1.33e-13 ***
## Treatment175          -23.7310     2.9857 659.0000  -7.948 8.23e-15 ***
## Treatment176          -17.4436     2.9857 659.0000  -5.842 8.09e-09 ***
## Treatment177          -23.0563     2.9857 659.0000  -7.722 4.26e-14 ***
## Treatment178          -22.8026     2.9857 659.0000  -7.637 7.84e-14 ***
## Treatment18           -19.5580     2.9857 659.0000  -6.550 1.16e-10 ***
## Treatment180          -19.8670     2.9857 659.0000  -6.654 6.01e-11 ***
## Treatment181          -23.1212     2.9857 659.0000  -7.744 3.65e-14 ***
## Treatment182          -23.1831     2.9857 659.0000  -7.765 3.14e-14 ***
## Treatment183          -22.9186     2.9857 659.0000  -7.676 5.94e-14 ***
## Treatment184          -21.7708     2.9857 659.0000  -7.292 8.81e-13 ***
## Treatment185          -23.3490     2.9857 659.0000  -7.820 2.10e-14 ***
## Treatment187          -23.5315     2.9857 659.0000  -7.881 1.34e-14 ***
## Treatment188          -22.4745     2.9857 659.0000  -7.527 1.71e-13 ***
## Treatment189          -17.0974     2.9857 659.0000  -5.726 1.56e-08 ***
## Treatment19           -16.5916     2.9857 659.0000  -5.557 3.99e-08 ***
## Treatment190          -20.0738     2.9857 659.0000  -6.723 3.86e-11 ***
## Treatment191          -24.5278     2.9857 659.0000  -8.215 1.12e-15 ***
## Treatment192          -23.2140     2.9857 659.0000  -7.775 2.91e-14 ***
## Treatment193          -23.3218     2.9857 659.0000  -7.811 2.24e-14 ***
## Treatment194          -16.6448     2.9857 659.0000  -5.575 3.62e-08 ***
## Treatment195          -21.1479     2.9857 659.0000  -7.083 3.63e-12 ***
## Treatment196          -23.5560     2.9857 659.0000  -7.889 1.27e-14 ***
## Treatment197          -25.1092     2.9857 659.0000  -8.410 2.54e-16 ***
## Treatment198          -22.6950     2.9857 659.0000  -7.601 1.01e-13 ***
```

```
## Treatment199          -24.0070     2.9857 659.0000  -8.041 4.15e-15 ***
## Treatment2            -20.6768     2.9857 659.0000  -6.925 1.04e-11 ***
## Treatment20           -18.7366     2.9857 659.0000  -6.275 6.32e-10 ***
## Treatment200          -23.0469     2.9857 659.0000  -7.719 4.36e-14 ***
## Treatment201          -23.2180     2.9857 659.0000  -7.776 2.88e-14 ***
## Treatment203          -17.5279     2.9857 659.0000  -5.871 6.89e-09 ***
## Treatment204            7.2552     2.9857 659.0000   2.430 0.015367 *
## Treatment206            2.0386     2.9857 659.0000   0.683 0.494983
## Treatment207          -20.7000     2.9857 659.0000  -6.933 9.85e-12 ***
## Treatment208          -17.6783     2.9857 659.0000  -5.921 5.15e-09 ***
## Treatment209          -21.7914     2.9857 659.0000  -7.298 8.40e-13 ***
## Treatment21           -21.9049     2.9857 659.0000  -7.336 6.47e-13 ***
## Treatment210          -19.4160     2.9857 659.0000  -6.503 1.56e-10 ***
## Treatment211           -9.8669     2.9857 659.0000  -3.305 0.001002 **
## Treatment212          -19.9195     2.9857 659.0000  -6.672 5.37e-11 ***
## Treatment213          -20.9319     2.9857 659.0000  -7.011 5.89e-12 ***
## Treatment214          -22.3391     2.9857 659.0000  -7.482 2.35e-13 ***
## Treatment215          -22.2894     2.9857 659.0000  -7.465 2.64e-13 ***
## Treatment216          -22.0432     2.9857 659.0000  -7.383 4.69e-13 ***
## Treatment217          -23.0857     2.9857 659.0000  -7.732 3.97e-14 ***
## Treatment218          -21.9736     2.9857 659.0000  -7.359 5.52e-13 ***
## Treatment219          -22.4752     2.9857 659.0000  -7.528 1.71e-13 ***
## Treatment22           -16.6017     2.9857 659.0000  -5.560 3.91e-08 ***
## Treatment220          -22.4776     2.9857 659.0000  -7.528 1.70e-13 ***
## Treatment221          -22.3850     2.9857 659.0000  -7.497 2.11e-13 ***
## Treatment222          -23.6983     2.9857 659.0000  -7.937 8.92e-15 ***
## Treatment223          -16.8680     2.5857 659.0000  -6.523 1.37e-10 ***
## Treatment224          -22.8426     2.9857 659.0000  -7.651 7.12e-14 ***
## Treatment225          -22.5622     2.9857 659.0000  -7.557 1.39e-13 ***
## Treatment226          -23.1652     2.9857 659.0000  -7.759 3.28e-14 ***
## Treatment227          -22.7919     2.9857 659.0000  -7.634 8.04e-14 ***
## Treatment228          -22.7009     2.9857 659.0000  -7.603 9.99e-14 ***
## Treatment229          -22.3999     2.9857 659.0000  -7.502 2.04e-13 ***
## Treatment23           -20.3971     2.9857 659.0000  -6.831 1.91e-11 ***
## Treatment230          -21.9422     2.9857 659.0000  -7.349 5.93e-13 ***
## Treatment231          -15.6104     2.9857 659.0000  -5.228 2.30e-07 ***
## Treatment234          -17.5323     2.9857 659.0000  -5.872 6.83e-09 ***
## Treatment235           -4.0884     2.9857 659.0000  -1.369 0.171370
## Treatment238          -18.5496     2.9857 659.0000  -6.213 9.23e-10 ***
## Treatment239           -4.8630     2.9857 659.0000  -1.629 0.103847
## Treatment24           -21.3848     2.9857 659.0000  -7.162 2.13e-12 ***
## Treatment240          -11.1941     2.9857 659.0000  -3.749 0.000193 ***
## Treatment243          -18.6702     2.9857 659.0000  -6.253 7.23e-10 ***
## Treatment245          -10.5657     2.9857 659.0000  -3.539 0.000430 ***
## Treatment246          -13.8533     2.9857 659.0000  -4.640 4.21e-06 ***
## Treatment247           -9.5342     2.9857 659.0000  -3.193 0.001474 **
## Treatment249          -11.1811     2.9857 659.0000  -3.745 0.000196 ***
## Treatment25           -21.5033     2.9857 659.0000  -7.202 1.63e-12 ***
## Treatment26           -22.2724     2.9857 659.0000  -7.460 2.75e-13 ***
## Treatment27           -16.1872     2.9857 659.0000  -5.422 8.30e-08 ***
## Treatment28           -14.5419     2.9857 659.0000  -4.870 1.40e-06 ***
## Treatment29            -8.8018     2.9857 659.0000  -2.948 0.003312 **
## Treatment3             10.9398     2.9857 659.0000   3.664 0.000268 ***
## Treatment30           -12.5137     2.9857 659.0000  -4.191 3.15e-05 ***
```

```
## Treatment32          -13.6398     2.9857 659.0000  -4.568 5.87e-06 ***
## Treatment34          -17.1457     2.9857 659.0000  -5.743 1.42e-08 ***
## Treatment36            4.1398     2.9857 659.0000   1.387 0.166061
## Treatment38          -14.5149     2.9857 659.0000  -4.861 1.46e-06 ***
## Treatment4           -21.1004     2.9857 659.0000  -7.067 4.04e-12 ***
## Treatment41          -15.5354     2.9857 659.0000  -5.203 2.62e-07 ***
## Treatment42           -9.6901     2.9857 659.0000  -3.245 0.001232 **
## Treatment43           -4.1700     2.9857 659.0000  -1.397 0.162995
## Treatment45          -16.5100     2.9857 659.0000  -5.530 4.63e-08 ***
## Treatment46          -21.5025     2.9857 659.0000  -7.202 1.63e-12 ***
## Treatment47          -22.4597     2.9857 659.0000  -7.522 1.77e-13 ***
## Treatment48          -15.5812     2.9857 659.0000  -5.219 2.42e-07 ***
## Treatment49          -22.5902     2.9857 659.0000  -7.566 1.30e-13 ***
## Treatment5           -19.4399     2.9857 659.0000  -6.511 1.48e-10 ***
## Treatment50            2.1230     2.9857 659.0000   0.711 0.477318
## Treatment52          -14.9538     2.9857 659.0000  -5.008 7.06e-07 ***
## Treatment53           -2.8255     2.9857 659.0000  -0.946 0.344326
## Treatment54            1.0374     2.9857 659.0000   0.347 0.728375
## Treatment55           -8.3344     2.9857 659.0000  -2.791 0.005400 **
## Treatment56           -6.8424     2.9857 659.0000  -2.292 0.022239 *
## Treatment57           -5.8446     2.9857 659.0000  -1.958 0.050710 .
## Treatment58          -20.8843     2.9857 659.0000  -6.995 6.55e-12 ***
## Treatment59            0.5553     2.5857 659.0000   0.215 0.830025
## Treatment6           -19.0915     2.9857 659.0000  -6.394 3.06e-10 ***
## Treatment61          -10.8890     2.9857 659.0000  -3.647 0.000286 ***
## Treatment62          -10.4994     2.9857 659.0000  -3.516 0.000467 ***
## Treatment63           -7.8132     2.9857 659.0000  -2.617 0.009078 **
## Treatment64            5.7740     2.9857 659.0000   1.934 0.053560 .
## Treatment65           -1.4091     2.9857 659.0000  -0.472 0.637113
## Treatment66           -5.9917     2.9857 659.0000  -2.007 0.045182 *
## Treatment67          -15.2302     2.9857 659.0000  -5.101 4.42e-07 ***
## Treatment68           -5.6076     2.9857 659.0000  -1.878 0.060804 .
## Treatment69           -1.8909     2.9857 659.0000  -0.633 0.526758
## Treatment7           -18.3980     2.9857 659.0000  -6.162 1.25e-09 ***
## Treatment70          -12.7171     2.9857 659.0000  -4.259 2.35e-05 ***
## Treatment71          -10.9495     2.9857 659.0000  -3.667 0.000265 ***
## Treatment72          -13.7063     2.9857 659.0000  -4.591 5.29e-06 ***
## Treatment73          -16.6507     2.9857 659.0000  -5.577 3.58e-08 ***
## Treatment74          -18.3053     2.9857 659.0000  -6.131 1.51e-09 ***
## Treatment75          -19.7189     2.9857 659.0000  -6.604 8.23e-11 ***
## Treatment76          -18.7190     2.9857 659.0000  -6.269 6.55e-10 ***
## Treatment77          -16.8549     2.9857 659.0000  -5.645 2.45e-08 ***
## Treatment78          -21.5000     2.9857 659.0000  -7.201 1.64e-12 ***
## Treatment79          -23.8033     2.9857 659.0000  -7.972 6.88e-15 ***
## Treatment8           -19.7742     2.9857 659.0000  -6.623 7.32e-11 ***
## Treatment80          -17.3594     2.9857 659.0000  -5.814 9.50e-09 ***
## Treatment81          -13.1774     2.9857 659.0000  -4.413 1.19e-05 ***
## Treatment82          -16.0393     2.9857 659.0000  -5.372 1.08e-07 ***
## Treatment83          -16.3290     2.9857 659.0000  -5.469 6.43e-08 ***
## Treatment84          -19.7435     2.9857 659.0000  -6.613 7.81e-11 ***
## Treatment85          -20.5346     2.9857 659.0000  -6.878 1.42e-11 ***
## Treatment86          -12.5914     2.9857 659.0000  -4.217 2.82e-05 ***
## Treatment87          -14.7294     2.9857 659.0000  -4.933 1.03e-06 ***
## Treatment88          -15.9760     2.9857 659.0000  -5.351 1.21e-07 ***
```

```
## Treatment89               -15.1178     2.9857 659.0000  -5.063 5.35e-07 ***
## Treatment9                -13.6355     2.9857 659.0000  -4.567 5.91e-06 ***
## Treatment90               -19.5662     2.9857 659.0000  -6.553 1.14e-10 ***
## Treatment91               -19.2722     2.9857 659.0000  -6.455 2.10e-10 ***
## Treatment92               -21.3903     2.9857 659.0000  -7.164 2.10e-12 ***
## Treatment93               -16.4227     2.9857 659.0000  -5.500 5.43e-08 ***
## Treatment94               -19.9551     2.9857 659.0000  -6.683 4.97e-11 ***
## Treatment95               -21.3849     2.9857 659.0000  -7.162 2.13e-12 ***
## Treatment97               -20.5758     2.9857 659.0000  -6.891 1.29e-11 ***
## Treatment98               -22.2764     2.9857 659.0000  -7.461 2.72e-13 ***
## Treatment99               -19.9009     2.9857 659.0000  -6.665 5.59e-11 ***
## TreatmentFluopyram         73.4603     2.9857 659.0000  24.604  < 2e-16 ***
## TreatmentUntreatedControl -23.6700     2.9857 659.0000  -7.928 9.56e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 217 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

8 ANOVA and LSMeans

```r
# ANOVA model
anova_mod <- aov(Mortality_percent ~ Treatment + Replication, data = Mortality)
summary(anova_mod)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Treatment   216  74887   346.7  19.443 <2e-16 ***
## Replication   3     52    17.4   0.975  0.404
## Residuals   656  11697    17.8
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Calculate estimated marginal means (LSMeans) with Benferroni adjustment
lsm <- emmeans(anova_mod, pairwise ~ Treatment, adjust = "bon")
```

9 Treatment vs Fluopyram Comparison

```r
Fluopyram <- contrast(lsm, "trt.vs.ctrl", ref = 1, adjust = "bon") # Compare each treatment to a refere
Fluopyram
```

```
##  contrast             estimate   SE  df t.ratio p.value
##  10 - 1                -22.527 2.99 656  -7.545  <.0001
```

```
##  100 - 1              -20.117 2.99 656  -6.737  <.0001
##  101 - 1              -21.726 2.99 656  -7.276  <.0001
##  103 - 1                1.255 2.99 656   0.420  1.0000
##  104 - 1              -23.673 2.99 656  -7.928  <.0001
##  105 - 1              -16.129 2.99 656  -5.402  <.0001
##  106 - 1              -23.662 2.99 656  -7.924  <.0001
##  107 - 1              -17.177 2.99 656  -5.753  <.0001
##  108 - 1              -20.149 2.99 656  -6.748  <.0001
##  11 - 1               -16.211 2.99 656  -5.429  <.0001
##  110 - 1              -21.246 2.99 656  -7.115  <.0001
##  111 - 1              -16.010 2.99 656  -5.362  <.0001
##  112 - 1              -22.441 2.99 656  -7.516  <.0001
##  113 - 1              -22.519 2.99 656  -7.542  <.0001
##  114 - 1              -24.528 2.99 656  -8.214  <.0001
##  115 - 1              -23.169 2.99 656  -7.760  <.0001
##  117 - 1              -21.601 2.99 656  -7.234  <.0001
##  118 - 1              -19.181 2.99 656  -6.424  <.0001
##  119 - 1              -23.923 2.99 656  -8.012  <.0001
##  12 - 1               -23.337 2.99 656  -7.816  <.0001
##  121 - 1              -15.507 2.99 656  -5.193  0.0001
##  122 - 1              -10.822 2.99 656  -3.624  0.0674
##  123 - 1              -15.451 2.99 656  -5.175  0.0001
##  124 - 1              -15.600 2.99 656  -5.225  0.0001
##  125 - 1              -13.702 2.99 656  -4.589  0.0012
##  126 - 1              -15.707 2.99 656  -5.260  <.0001
##  128 - 1              -23.317 2.99 656  -7.809  <.0001
##  129 - 1              -15.539 2.99 656  -5.204  0.0001
##  13 - 1               -23.001 2.99 656  -7.703  <.0001
##  130 - 1              -13.950 2.99 656  -4.672  0.0008
##  131 - 1              -23.810 2.99 656  -7.974  <.0001
##  133 - 1              -20.513 2.99 656  -6.870  <.0001
##  134 - 1              -15.091 2.99 656  -5.054  0.0001
##  135 - 1              -24.609 2.99 656  -8.242  <.0001
##  136 - 1              -15.832 2.99 656  -5.302  <.0001
##  137 - 1              -11.792 2.99 656  -3.949  0.0188
##  138 - 1              -16.624 2.99 656  -5.567  <.0001
##  139 - 1              -15.856 2.99 656  -5.310  <.0001
##  14 - 1                -6.813 2.99 656  -2.282  1.0000
##  141 - 1              -21.253 2.99 656  -7.118  <.0001
##  143 - 1              -23.280 2.99 656  -7.797  <.0001
##  146 - 1              -19.475 2.99 656  -6.522  <.0001
##  147 - 1              -23.835 2.99 656  -7.983  <.0001
##  148 - 1              -22.525 2.99 656  -7.544  <.0001
##  149 - 1              -17.455 2.99 656  -5.846  <.0001
##  15 - 1               -22.028 2.99 656  -7.377  <.0001
##  150 - 1              -22.584 2.99 656  -7.564  <.0001
##  151 - 1              -19.280 2.99 656  -6.457  <.0001
##  152 - 1              -12.856 2.99 656  -4.306  0.0041
##  153 - 1              -20.625 2.99 656  -6.907  <.0001
##  154 - 1              -20.530 2.99 656  -6.876  <.0001
##  155 - 1              -20.292 2.99 656  -6.796  <.0001
##  156 - 1              -22.440 2.99 656  -7.515  <.0001
##  157 - 1              -12.815 2.99 656  -4.292  0.0044
##  158 - 1              -24.098 2.99 656  -8.071  <.0001
```

```
##  159 - 1              -22.958 2.99 656  -7.689  <.0001
##  16 - 1               -20.342 2.99 656  -6.813  <.0001
##  160 - 1              -21.768 2.99 656  -7.290  <.0001
##  162 - 1              -23.814 2.99 656  -7.975  <.0001
##  163 - 1              -21.881 2.99 656  -7.328  <.0001
##  164 - 1              -24.047 2.99 656  -8.053  <.0001
##  165 - 1              -23.493 2.99 656  -7.868  <.0001
##  166 - 1              -24.166 2.99 656  -8.093  <.0001
##  167 - 1              -18.941 2.99 656  -6.343  <.0001
##  168 - 1              -21.009 2.99 656  -7.036  <.0001
##  169 - 1              -21.388 2.99 656  -7.163  <.0001
##  17 - 1               -19.758 2.99 656  -6.617  <.0001
##  170 - 1              -20.406 2.99 656  -6.834  <.0001
##  171 - 1              -23.102 2.99 656  -7.737  <.0001
##  172 - 1              -21.823 2.99 656  -7.308  <.0001
##  174 - 1              -22.582 2.99 656  -7.563  <.0001
##  175 - 1              -23.731 2.99 656  -7.948  <.0001
##  176 - 1              -17.444 2.99 656  -5.842  <.0001
##  177 - 1              -23.056 2.99 656  -7.722  <.0001
##  178 - 1              -22.803 2.99 656  -7.637  <.0001
##  18 - 1               -19.558 2.99 656  -6.550  <.0001
##  180 - 1              -19.867 2.99 656  -6.654  <.0001
##  181 - 1              -23.121 2.99 656  -7.743  <.0001
##  182 - 1              -23.183 2.99 656  -7.764  <.0001
##  183 - 1              -22.919 2.99 656  -7.676  <.0001
##  184 - 1              -21.771 2.99 656  -7.291  <.0001
##  185 - 1              -23.349 2.99 656  -7.820  <.0001
##  187 - 1              -23.532 2.99 656  -7.881  <.0001
##  188 - 1              -22.475 2.99 656  -7.527  <.0001
##  189 - 1              -17.097 2.99 656  -5.726  <.0001
##  19 - 1               -16.592 2.99 656  -5.557  <.0001
##  190 - 1              -20.074 2.99 656  -6.723  <.0001
##  191 - 1              -24.528 2.99 656  -8.214  <.0001
##  192 - 1              -23.214 2.99 656  -7.774  <.0001
##  193 - 1              -23.322 2.99 656  -7.811  <.0001
##  194 - 1              -16.645 2.99 656  -5.574  <.0001
##  195 - 1              -21.148 2.99 656  -7.083  <.0001
##  196 - 1              -23.556 2.99 656  -7.889  <.0001
##  197 - 1              -25.109 2.99 656  -8.409  <.0001
##  198 - 1              -22.695 2.99 656  -7.601  <.0001
##  199 - 1              -24.007 2.99 656  -8.040  <.0001
##  2 - 1                -20.677 2.99 656  -6.925  <.0001
##  20 - 1               -18.737 2.99 656  -6.275  <.0001
##  200 - 1              -23.047 2.99 656  -7.719  <.0001
##  201 - 1              -23.218 2.99 656  -7.776  <.0001
##  203 - 1              -17.528 2.99 656  -5.870  <.0001
##  204 - 1                7.255 2.99 656   2.430  1.0000
##  206 - 1                2.039 2.99 656   0.683  1.0000
##  207 - 1              -20.700 2.99 656  -6.933  <.0001
##  208 - 1              -17.678 2.99 656  -5.921  <.0001
##  209 - 1              -21.791 2.99 656  -7.298  <.0001
##  21 - 1               -21.905 2.99 656  -7.336  <.0001
##  210 - 1              -19.416 2.99 656  -6.503  <.0001
##  211 - 1               -9.867 2.99 656  -3.304  0.2167
```

```
##  212 - 1              -19.920 2.99 656  -6.671  <.0001
##  213 - 1              -20.932 2.99 656  -7.010  <.0001
##  214 - 1              -22.339 2.99 656  -7.481  <.0001
##  215 - 1              -22.289 2.99 656  -7.465  <.0001
##  216 - 1              -22.043 2.99 656  -7.382  <.0001
##  217 - 1              -23.086 2.99 656  -7.732  <.0001
##  218 - 1              -21.974 2.99 656  -7.359  <.0001
##  219 - 1              -22.475 2.99 656  -7.527  <.0001
##  22 - 1               -16.602 2.99 656  -5.560  <.0001
##  220 - 1              -22.478 2.99 656  -7.528  <.0001
##  221 - 1              -22.385 2.99 656  -7.497  <.0001
##  222 - 1              -23.698 2.99 656  -7.937  <.0001
##  223 - 1              -16.868 2.59 656  -6.523  <.0001
##  224 - 1              -22.843 2.99 656  -7.650  <.0001
##  225 - 1              -22.562 2.99 656  -7.556  <.0001
##  226 - 1              -23.165 2.99 656  -7.758  <.0001
##  227 - 1              -22.792 2.99 656  -7.633  <.0001
##  228 - 1              -22.701 2.99 656  -7.603  <.0001
##  229 - 1              -22.400 2.99 656  -7.502  <.0001
##  23 - 1               -20.397 2.99 656  -6.831  <.0001
##  230 - 1              -21.942 2.99 656  -7.349  <.0001
##  231 - 1              -15.610 2.99 656  -5.228  <.0001
##  234 - 1              -17.532 2.99 656  -5.872  <.0001
##  235 - 1               -4.088 2.99 656  -1.369  1.0000
##  238 - 1              -18.550 2.99 656  -6.212  <.0001
##  239 - 1               -4.863 2.99 656  -1.629  1.0000
##  24 - 1               -21.385 2.99 656  -7.162  <.0001
##  240 - 1              -11.194 2.99 656  -3.749  0.0417
##  243 - 1              -18.670 2.99 656  -6.253  <.0001
##  245 - 1              -10.566 2.99 656  -3.539  0.0930
##  246 - 1              -13.853 2.99 656  -4.640  0.0009
##  247 - 1               -9.534 2.99 656  -3.193  0.3186
##  249 - 1              -11.181 2.99 656  -3.745  0.0425
##  25 - 1               -21.503 2.99 656  -7.202  <.0001
##  26 - 1               -22.272 2.99 656  -7.459  <.0001
##  27 - 1               -16.187 2.99 656  -5.421  <.0001
##  28 - 1               -14.542 2.99 656  -4.870  0.0003
##  29 - 1                -8.802 2.99 656  -2.948  0.7159
##  3 - 1                 10.940 2.99 656   3.664  0.0580
##  30 - 1               -12.514 2.99 656  -4.191  0.0068
##  32 - 1               -13.640 2.99 656  -4.568  0.0013
##  34 - 1               -17.146 2.99 656  -5.742  <.0001
##  36 - 1                 4.140 2.99 656   1.386  1.0000
##  38 - 1               -14.515 2.99 656  -4.861  0.0003
##  4 - 1                -21.100 2.99 656  -7.067  <.0001
##  41 - 1               -15.535 2.99 656  -5.203  0.0001
##  42 - 1                -9.690 2.99 656  -3.245  0.2663
##  43 - 1                -4.170 2.99 656  -1.397  1.0000
##  45 - 1               -16.510 2.99 656  -5.529  <.0001
##  46 - 1               -21.502 2.99 656  -7.201  <.0001
##  47 - 1               -22.460 2.99 656  -7.522  <.0001
##  48 - 1               -15.581 2.99 656  -5.218  0.0001
##  49 - 1               -22.590 2.99 656  -7.566  <.0001
##  5 - 1                -19.440 2.99 656  -6.511  <.0001
```

```
##  50 - 1                  2.123 2.99 656   0.711  1.0000
##  52 - 1                -14.954 2.99 656  -5.008  0.0002
##  53 - 1                 -2.826 2.99 656  -0.946  1.0000
##  54 - 1                  1.037 2.99 656   0.347  1.0000
##  55 - 1                 -8.334 2.99 656  -2.791  1.0000
##  56 - 1                 -6.842 2.99 656  -2.292  1.0000
##  57 - 1                 -5.845 2.99 656  -1.957  1.0000
##  58 - 1                -20.884 2.99 656  -6.994  <.0001
##  59 - 1                  0.555 2.59 656   0.215  1.0000
##  6 - 1                 -19.092 2.99 656  -6.394  <.0001
##  61 - 1                -10.889 2.99 656  -3.647  0.0619
##  62 - 1                -10.499 2.99 656  -3.516  0.1010
##  63 - 1                 -7.813 2.99 656  -2.617  1.0000
##  64 - 1                  5.774 2.99 656   1.934  1.0000
##  65 - 1                 -1.409 2.99 656  -0.472  1.0000
##  66 - 1                 -5.992 2.99 656  -2.007  1.0000
##  67 - 1                -15.230 2.99 656  -5.101  0.0001
##  68 - 1                 -5.608 2.99 656  -1.878  1.0000
##  69 - 1                 -1.891 2.99 656  -0.633  1.0000
##  7 - 1                 -18.398 2.99 656  -6.162  <.0001
##  70 - 1                -12.717 2.99 656  -4.259  0.0051
##  71 - 1                -10.950 2.99 656  -3.667  0.0573
##  72 - 1                -13.706 2.99 656  -4.590  0.0011
##  73 - 1                -16.651 2.99 656  -5.576  <.0001
##  74 - 1                -18.305 2.99 656  -6.131  <.0001
##  75 - 1                -19.719 2.99 656  -6.604  <.0001
##  76 - 1                -18.719 2.99 656  -6.269  <.0001
##  77 - 1                -16.855 2.99 656  -5.645  <.0001
##  78 - 1                -21.500 2.99 656  -7.200  <.0001
##  79 - 1                -23.803 2.99 656  -7.972  <.0001
##  8 - 1                 -19.774 2.99 656  -6.622  <.0001
##  80 - 1                -17.359 2.99 656  -5.814  <.0001
##  81 - 1                -13.177 2.99 656  -4.413  0.0026
##  82 - 1                -16.039 2.99 656  -5.372  <.0001
##  83 - 1                -16.329 2.99 656  -5.469  <.0001
##  84 - 1                -19.744 2.99 656  -6.612  <.0001
##  85 - 1                -20.535 2.99 656  -6.877  <.0001
##  86 - 1                -12.591 2.99 656  -4.217  0.0061
##  87 - 1                -14.729 2.99 656  -4.933  0.0002
##  88 - 1                -15.976 2.99 656  -5.350  <.0001
##  89 - 1                -15.118 2.99 656  -5.063  0.0001
##  9 - 1                 -13.635 2.99 656  -4.567  0.0013
##  90 - 1                -19.566 2.99 656  -6.553  <.0001
##  91 - 1                -19.272 2.99 656  -6.454  <.0001
##  92 - 1                -21.390 2.99 656  -7.164  <.0001
##  93 - 1                -16.423 2.99 656  -5.500  <.0001
##  94 - 1                -19.955 2.99 656  -6.683  <.0001
##  95 - 1                -21.385 2.99 656  -7.162  <.0001
##  97 - 1                -20.576 2.99 656  -6.891  <.0001
##  98 - 1                -22.276 2.99 656  -7.460  <.0001
##  99 - 1                -19.901 2.99 656  -6.665  <.0001
##  Fluopyram - 1          73.460 2.99 656  24.602  <.0001
##  UntreatedControl - 1  -23.670 2.99 656  -7.927  <.0001
##
```

```
## Results are averaged over the levels of: Replication
## P value adjustment: bonferroni method for 216 tests
```

10 Treatment vs Untreated Control comparison

```r
untreatedcontrol <- contrast(lsm, "trt.vs.ctrl", ref = 217, adjust = "bon") # Compare each treatment to
untreatedcontrol
```

```
##  contrast                     estimate   SE  df t.ratio p.value
##  1 - UntreatedControl         23.67005 2.99 656   7.927  <.0001
##  10 - UntreatedControl         1.14255 2.99 656   0.383  1.0000
##  100 - UntreatedControl        3.55301 2.99 656   1.190  1.0000
##  101 - UntreatedControl        1.94368 2.99 656   0.651  1.0000
##  103 - UntreatedControl       24.92527 2.99 656   8.348  <.0001
##  104 - UntreatedControl       -0.00301 2.99 656  -0.001  1.0000
##  105 - UntreatedControl        7.54096 2.99 656   2.526  1.0000
##  106 - UntreatedControl        0.00833 2.99 656   0.003  1.0000
##  107 - UntreatedControl        6.49295 2.99 656   2.175  1.0000
##  108 - UntreatedControl        3.52117 2.99 656   1.179  1.0000
##  11 - UntreatedControl         7.45919 2.99 656   2.498  1.0000
##  110 - UntreatedControl        2.42386 2.99 656   0.812  1.0000
##  111 - UntreatedControl        7.66007 2.99 656   2.565  1.0000
##  112 - UntreatedControl        1.22937 2.99 656   0.412  1.0000
##  113 - UntreatedControl        1.15113 2.99 656   0.386  1.0000
##  114 - UntreatedControl       -0.85775 2.99 656  -0.287  1.0000
##  115 - UntreatedControl        0.50080 2.99 656   0.168  1.0000
##  117 - UntreatedControl        2.06951 2.99 656   0.693  1.0000
##  118 - UntreatedControl        4.48870 2.99 656   1.503  1.0000
##  119 - UntreatedControl       -0.25326 2.99 656  -0.085  1.0000
##  12 - UntreatedControl         0.33340 2.99 656   0.112  1.0000
##  121 - UntreatedControl        8.16347 2.99 656   2.734  1.0000
##  122 - UntreatedControl       12.84813 2.99 656   4.303  0.0042
##  123 - UntreatedControl        8.21941 2.99 656   2.753  1.0000
##  124 - UntreatedControl        8.06988 2.99 656   2.703  1.0000
##  125 - UntreatedControl        9.96830 2.99 656   3.338  0.1923
##  126 - UntreatedControl        7.96326 2.99 656   2.667  1.0000
##  128 - UntreatedControl        0.35260 2.99 656   0.118  1.0000
##  129 - UntreatedControl        8.13084 2.99 656   2.723  1.0000
##  13 - UntreatedControl         0.66919 2.99 656   0.224  1.0000
##  130 - UntreatedControl        9.71994 2.99 656   3.255  0.2573
##  131 - UntreatedControl       -0.14022 2.99 656  -0.047  1.0000
##  133 - UntreatedControl        3.15702 2.99 656   1.057  1.0000
##  134 - UntreatedControl        8.57955 2.99 656   2.873  0.9057
##  135 - UntreatedControl       -0.93914 2.99 656  -0.315  1.0000
##  136 - UntreatedControl        7.83854 2.99 656   2.625  1.0000
##  137 - UntreatedControl       11.87757 2.99 656   3.978  0.0167
##  138 - UntreatedControl        7.04617 2.99 656   2.360  1.0000
##  139 - UntreatedControl        7.81446 2.99 656   2.617  1.0000
##  14 - UntreatedControl        16.85732 2.99 656   5.646  <.0001
##  141 - UntreatedControl        2.41702 2.99 656   0.809  1.0000
##  143 - UntreatedControl        0.39012 2.99 656   0.131  1.0000
```

```
##  146 - UntreatedControl        4.19530 2.99 656   1.405  1.0000
##  147 - UntreatedControl       -0.16513 2.99 656  -0.055  1.0000
##  148 - UntreatedControl        1.14473 2.99 656   0.383  1.0000
##  149 - UntreatedControl        6.21503 2.99 656   2.081  1.0000
##  15 - UntreatedControl         1.64171 2.99 656   0.550  1.0000
##  150 - UntreatedControl        1.08569 2.99 656   0.364  1.0000
##  151 - UntreatedControl        4.39027 2.99 656   1.470  1.0000
##  152 - UntreatedControl       10.81406 2.99 656   3.622  0.0681
##  153 - UntreatedControl        3.04554 2.99 656   1.020  1.0000
##  154 - UntreatedControl        3.14035 2.99 656   1.052  1.0000
##  155 - UntreatedControl        3.37829 2.99 656   1.131  1.0000
##  156 - UntreatedControl        1.22968 2.99 656   0.412  1.0000
##  157 - UntreatedControl       10.85542 2.99 656   3.636  0.0646
##  158 - UntreatedControl       -0.42812 2.99 656  -0.143  1.0000
##  159 - UntreatedControl        0.71241 2.99 656   0.239  1.0000
##  16 - UntreatedControl         3.32758 2.99 656   1.114  1.0000
##  160 - UntreatedControl        1.90187 2.99 656   0.637  1.0000
##  162 - UntreatedControl       -0.14346 2.99 656  -0.048  1.0000
##  163 - UntreatedControl        1.78867 2.99 656   0.599  1.0000
##  164 - UntreatedControl       -0.37698 2.99 656  -0.126  1.0000
##  165 - UntreatedControl        0.17693 2.99 656   0.059  1.0000
##  166 - UntreatedControl       -0.49575 2.99 656  -0.166  1.0000
##  167 - UntreatedControl        4.72905 2.99 656   1.584  1.0000
##  168 - UntreatedControl        2.66138 2.99 656   0.891  1.0000
##  169 - UntreatedControl        2.28203 2.99 656   0.764  1.0000
##  17 - UntreatedControl         3.91254 2.99 656   1.310  1.0000
##  170 - UntreatedControl        3.26441 2.99 656   1.093  1.0000
##  171 - UntreatedControl        0.56772 2.99 656   0.190  1.0000
##  172 - UntreatedControl        1.84749 2.99 656   0.619  1.0000
##  174 - UntreatedControl        1.08820 2.99 656   0.364  1.0000
##  175 - UntreatedControl       -0.06094 2.99 656  -0.020  1.0000
##  176 - UntreatedControl        6.22649 2.99 656   2.085  1.0000
##  177 - UntreatedControl        0.61370 2.99 656   0.206  1.0000
##  178 - UntreatedControl        0.86740 2.99 656   0.290  1.0000
##  18 - UntreatedControl         4.11202 2.99 656   1.377  1.0000
##  180 - UntreatedControl        3.80304 2.99 656   1.274  1.0000
##  181 - UntreatedControl        0.54885 2.99 656   0.184  1.0000
##  182 - UntreatedControl        0.48697 2.99 656   0.163  1.0000
##  183 - UntreatedControl        0.75147 2.99 656   0.252  1.0000
##  184 - UntreatedControl        1.89922 2.99 656   0.636  1.0000
##  185 - UntreatedControl        0.32106 2.99 656   0.108  1.0000
##  187 - UntreatedControl        0.13851 2.99 656   0.046  1.0000
##  188 - UntreatedControl        1.19551 2.99 656   0.400  1.0000
##  189 - UntreatedControl        6.57263 2.99 656   2.201  1.0000
##  19 - UntreatedControl         7.07844 2.99 656   2.371  1.0000
##  190 - UntreatedControl        3.59625 2.99 656   1.204  1.0000
##  191 - UntreatedControl       -0.85775 2.99 656  -0.287  1.0000
##  192 - UntreatedControl        0.45604 2.99 656   0.153  1.0000
##  193 - UntreatedControl        0.34822 2.99 656   0.117  1.0000
##  194 - UntreatedControl        7.02524 2.99 656   2.353  1.0000
##  195 - UntreatedControl        2.52218 2.99 656   0.845  1.0000
##  196 - UntreatedControl        0.11401 2.99 656   0.038  1.0000
##  197 - UntreatedControl       -1.43914 2.99 656  -0.482  1.0000
##  198 - UntreatedControl        0.97510 2.99 656   0.327  1.0000
```

```
##  199 - UntreatedControl       -0.33692 2.99 656  -0.113  1.0000
##  2 - UntreatedControl          2.99326 2.99 656   1.002  1.0000
##  20 - UntreatedControl         4.93349 2.99 656   1.652  1.0000
##  200 - UntreatedControl        0.62318 2.99 656   0.209  1.0000
##  201 - UntreatedControl        0.45204 2.99 656   0.151  1.0000
##  203 - UntreatedControl        6.14218 2.99 656   2.057  1.0000
##  204 - UntreatedControl       30.92529 2.99 656  10.357  <.0001
##  206 - UntreatedControl       25.70867 2.99 656   8.610  <.0001
##  207 - UntreatedControl        2.97000 2.99 656   0.995  1.0000
##  208 - UntreatedControl        5.99174 2.99 656   2.007  1.0000
##  209 - UntreatedControl        1.87869 2.99 656   0.629  1.0000
##  21 - UntreatedControl         1.76519 2.99 656   0.591  1.0000
##  210 - UntreatedControl        4.25401 2.99 656   1.425  1.0000
##  211 - UntreatedControl       13.80310 2.99 656   4.623  0.0010
##  212 - UntreatedControl        3.75051 2.99 656   1.256  1.0000
##  213 - UntreatedControl        2.73818 2.99 656   0.917  1.0000
##  214 - UntreatedControl        1.33098 2.99 656   0.446  1.0000
##  215 - UntreatedControl        1.38068 2.99 656   0.462  1.0000
##  216 - UntreatedControl        1.62681 2.99 656   0.545  1.0000
##  217 - UntreatedControl        0.58437 2.99 656   0.196  1.0000
##  218 - UntreatedControl        1.69648 2.99 656   0.568  1.0000
##  219 - UntreatedControl        1.19482 2.99 656   0.400  1.0000
##  22 - UntreatedControl         7.06831 2.99 656   2.367  1.0000
##  220 - UntreatedControl        1.19247 2.99 656   0.399  1.0000
##  221 - UntreatedControl        1.28507 2.99 656   0.430  1.0000
##  222 - UntreatedControl       -0.02822 2.99 656  -0.009  1.0000
##  223 - UntreatedControl        6.80205 2.59 656   2.630  1.0000
##  224 - UntreatedControl        0.82744 2.99 656   0.277  1.0000
##  225 - UntreatedControl        1.10789 2.99 656   0.371  1.0000
##  226 - UntreatedControl        0.50489 2.99 656   0.169  1.0000
##  227 - UntreatedControl        0.87814 2.99 656   0.294  1.0000
##  228 - UntreatedControl        0.96915 2.99 656   0.325  1.0000
##  229 - UntreatedControl        1.27017 2.99 656   0.425  1.0000
##  23 - UntreatedControl         3.27292 2.99 656   1.096  1.0000
##  230 - UntreatedControl        1.72787 2.99 656   0.579  1.0000
##  231 - UntreatedControl        8.05964 2.99 656   2.699  1.0000
##  234 - UntreatedControl        6.13773 2.99 656   2.056  1.0000
##  235 - UntreatedControl       19.58165 2.99 656   6.558  <.0001
##  238 - UntreatedControl        5.12040 2.99 656   1.715  1.0000
##  239 - UntreatedControl       18.80706 2.99 656   6.299  <.0001
##  24 - UntreatedControl         2.28526 2.99 656   0.765  1.0000
##  240 - UntreatedControl       12.47591 2.99 656   4.178  0.0072
##  243 - UntreatedControl        4.99987 2.99 656   1.674  1.0000
##  245 - UntreatedControl       13.10430 2.99 656   4.389  0.0029
##  246 - UntreatedControl        9.81677 2.99 656   3.288  0.2298
##  247 - UntreatedControl       14.13581 2.99 656   4.734  0.0006
##  249 - UntreatedControl       12.48890 2.99 656   4.183  0.0071
##  25 - UntreatedControl         2.16673 2.99 656   0.726  1.0000
##  26 - UntreatedControl         1.39767 2.99 656   0.468  1.0000
##  27 - UntreatedControl         7.48281 2.99 656   2.506  1.0000
##  28 - UntreatedControl         9.12811 2.99 656   3.057  0.5024
##  29 - UntreatedControl        14.86820 2.99 656   4.979  0.0002
##  3 - UntreatedControl         34.60988 2.99 656  11.591  <.0001
##  30 - UntreatedControl        11.15634 2.99 656   3.736  0.0438
```

```
##  32 - UntreatedControl        10.03029 2.99 656   3.359  0.1786
##  34 - UntreatedControl         6.52432 2.99 656   2.185  1.0000
##  36 - UntreatedControl        27.80980 2.99 656   9.314  <.0001
##  38 - UntreatedControl         9.15511 2.99 656   3.066  0.4877
##  4 - UntreatedControl          2.56964 2.99 656   0.861  1.0000
##  41 - UntreatedControl         8.13469 2.99 656   2.724  1.0000
##  42 - UntreatedControl        13.97996 2.99 656   4.682  0.0007
##  43 - UntreatedControl        19.50006 2.99 656   6.531  <.0001
##  45 - UntreatedControl         7.16002 2.99 656   2.398  1.0000
##  46 - UntreatedControl         2.16756 2.99 656   0.726  1.0000
##  47 - UntreatedControl         1.21037 2.99 656   0.405  1.0000
##  48 - UntreatedControl         8.08885 2.99 656   2.709  1.0000
##  49 - UntreatedControl         1.07985 2.99 656   0.362  1.0000
##  5 - UntreatedControl          4.23014 2.99 656   1.417  1.0000
##  50 - UntreatedControl        25.79300 2.99 656   8.638  <.0001
##  52 - UntreatedControl         8.71629 2.99 656   2.919  0.7842
##  53 - UntreatedControl        20.84454 2.99 656   6.981  <.0001
##  54 - UntreatedControl        24.70740 2.99 656   8.275  <.0001
##  55 - UntreatedControl        15.33568 2.99 656   5.136  0.0001
##  56 - UntreatedControl        16.82767 2.99 656   5.636  <.0001
##  57 - UntreatedControl        17.82543 2.99 656   5.970  <.0001
##  58 - UntreatedControl         2.78577 2.99 656   0.933  1.0000
##  59 - UntreatedControl        24.22534 2.59 656   9.368  <.0001
##  6 - UntreatedControl          4.57852 2.99 656   1.533  1.0000
##  61 - UntreatedControl        12.78102 2.99 656   4.280  0.0046
##  62 - UntreatedControl        13.17069 2.99 656   4.411  0.0026
##  63 - UntreatedControl        15.85684 2.99 656   5.311  <.0001
##  64 - UntreatedControl        29.44401 2.99 656   9.861  <.0001
##  65 - UntreatedControl        22.26090 2.99 656   7.455  <.0001
##  66 - UntreatedControl        17.67835 2.99 656   5.921  <.0001
##  67 - UntreatedControl         8.43982 2.99 656   2.827  1.0000
##  68 - UntreatedControl        18.06241 2.99 656   6.049  <.0001
##  69 - UntreatedControl        21.77918 2.99 656   7.294  <.0001
##  7 - UntreatedControl          5.27207 2.99 656   1.766  1.0000
##  70 - UntreatedControl        10.95291 2.99 656   3.668  0.0570
##  71 - UntreatedControl        12.72052 2.99 656   4.260  0.0051
##  72 - UntreatedControl         9.96374 2.99 656   3.337  0.1933
##  73 - UntreatedControl         7.01938 2.99 656   2.351  1.0000
##  74 - UntreatedControl         5.36475 2.99 656   1.797  1.0000
##  75 - UntreatedControl         3.95112 2.99 656   1.323  1.0000
##  76 - UntreatedControl         4.95109 2.99 656   1.658  1.0000
##  77 - UntreatedControl         6.81514 2.99 656   2.282  1.0000
##  78 - UntreatedControl         2.17010 2.99 656   0.727  1.0000
##  79 - UntreatedControl        -0.13323 2.99 656  -0.045  1.0000
##  8 - UntreatedControl          3.89589 2.99 656   1.305  1.0000
##  80 - UntreatedControl         6.31068 2.99 656   2.113  1.0000
##  81 - UntreatedControl        10.49267 2.99 656   3.514  0.1019
##  82 - UntreatedControl         7.63077 2.99 656   2.556  1.0000
##  83 - UntreatedControl         7.34104 2.99 656   2.459  1.0000
##  84 - UntreatedControl         3.92655 2.99 656   1.315  1.0000
##  85 - UntreatedControl         3.13548 2.99 656   1.050  1.0000
##  86 - UntreatedControl        11.07865 2.99 656   3.710  0.0485
##  87 - UntreatedControl         8.94065 2.99 656   2.994  0.6166
##  88 - UntreatedControl         7.69404 2.99 656   2.577  1.0000
```

```
##  89 - UntreatedControl         8.55227 2.99 656   2.864  0.9319
##  9 - UntreatedControl         10.03459 2.99 656   3.361  0.1777
##  90 - UntreatedControl         4.10381 2.99 656   1.374  1.0000
##  91 - UntreatedControl         4.39782 2.99 656   1.473  1.0000
##  92 - UntreatedControl         2.27979 2.99 656   0.764  1.0000
##  93 - UntreatedControl         7.24735 2.99 656   2.427  1.0000
##  94 - UntreatedControl         3.71492 2.99 656   1.244  1.0000
##  95 - UntreatedControl         2.28519 2.99 656   0.765  1.0000
##  97 - UntreatedControl         3.09421 2.99 656   1.036  1.0000
##  98 - UntreatedControl         1.39366 2.99 656   0.467  1.0000
##  99 - UntreatedControl         3.76919 2.99 656   1.262  1.0000
##  Fluopyram - UntreatedControl 97.13030 2.99 656  32.529  <.0001
##
## Results are averaged over the levels of: Replication
## P value adjustment: bonferroni method for 216 tests
```